

Project Design Phase

Proposed Solution Template

Date	18th June 2026
Team ID	
Project Name	MediCareBook (Book a Doctor)
Maximum Marks	2 Marks

Proposed Solution Evaluation Matrix:

Problem Statement	Clinic scheduling systems suffer from scheduling collisions, listing of unverified doctor credentials, and a lack of role security constraints, leaving confidential patient records open to unauthorized system views.
Idea / Solution Description	A MERN stack (MongoDB, Express, React, Node.js) web application following the MVC controller design pattern. It provides patient doctor specialization lookups, doctor dashboard approval queues, and administrator vetting tools.
Novelty / Uniqueness	Strictly validates request routes via custom Role-Based Access Control (RBAC) server middleware. Features an administrative gatekeeping queue: newly registered doctor profiles are suppressed from directories until validated and toggled to approved status by the administrator.
Social Impact / Customer Satisfaction	Builds clinical safety and user trust by highlighting approved doctors with a "Verified" badge. Shifting consultations to organized hourly schedules eliminates long clinic wait queues. Restricts private medical histories so they are only visible to the patient and their consulting practitioner.
Business Model (Revenue Model)	• Premium Placement Subscriptions: Verified doctors pay a monthly fee to index at the top of category searches. • Booking Convenience Fees: A nominal transactional processing charge per appointment scheduling. • Enterprise White-Label Licensing: Licensing the booking dashboard module directly to clinical networks.
Scalability of the Solution	• Presentation Tier: Bundled using React Vite, deploying easily on edge delivery networks. • Application Tier: Stateless server APIs can be cloned behind load balancers.

- **Database Tier:** Horizontal database cluster scaling via MongoDB Atlas sharded shards.